

Optimal Execution with Alpha

BayQuant

From Liquidation to Alpha

A *Primer on Optimal Execution* solved the Almgren–Chriss liquidation problem [1] with no directional view: the price drift α was set to zero. Here the trader has a view. Holding a share earns an expected return $\alpha(t)$, and the case of interest is an earnings event, whose alpha arrives in a single instant. The optimal schedule turns out to be the liquidation curve plus a Green’s-function response to the event. Adding the event’s own jump variance then caps the position carried into the print. The last section compares the model with the round-trip version in Paleologo [2].

The analysis is in continuous time. The discrete grid of the primer returns only to derive the optimality equation and to check the closed forms numerically. In continuous time, permanent impact contributes $\frac{1}{2}\gamma(x(T)^2 - x(0)^2)$ whatever the schedule, so only the temporary impact η shapes it.

The Objective

The trader holds $x(t)$ shares and must go from $x(0) = X$ to $x(T) = 0$. At every instant three things happen:

- trading costs $\eta\dot{x}^2$, quadratic in speed, so trading twice as fast costs four times as much;
- holding costs risk $\lambda\sigma^2x^2$;
- holding earns alpha $\alpha(t)x$.

The trader chooses the whole path to minimize

$$J[x] = \int_0^T [\eta\dot{x}^2 + \lambda\sigma^2x^2 - \alpha(t)x] dt. \quad (1)$$

This is the primer’s $E + \lambda V$ with the drift switched back on: every share held through a drift α earns it, which lowers the expected shortfall by $\int \alpha(t)x(t) dt$.

The Optimality Equation

To minimize a function of many variables, set every partial derivative to zero. Split $[0, T]$ into N buckets of length τ , with holdings $x_1, \dots, x_{N-1}$ free and the endpoints $x_0 = X$ and $x_N = 0$ fixed. The objective becomes

$$J = \sum_k \left[\frac{\eta}{\tau} (x_{k-1} - x_k)^2 + \lambda\sigma^2\tau x_k^2 - \alpha_k\tau x_k \right].$$

A single holding x_k appears in exactly four places: the trade into bucket k , the trade out of it, its own risk and its own alpha. Setting the derivative to zero,

$$\frac{2\eta}{\tau} (2x_k - x_{k-1} - x_{k+1}) + 2\lambda\sigma^2\tau x_k - \alpha_k\tau = 0,$$

and dividing by $2\eta\tau$,

$$\frac{x_{k+1} - 2x_k + x_{k-1}}{\tau^2} = \kappa^2 x_k - \frac{\alpha_k}{2\eta}, \quad \kappa^2 = \frac{\lambda\sigma^2}{\eta}.$$

The left side is the change in trading speed, from $(x_k - x_{k-1})/\tau$ into bucket k to $(x_{k+1} - x_k)/\tau$ out of it, divided by τ : a discrete acceleration. As $\tau \rightarrow 0$ it becomes

$$\ddot{x} - \kappa^2 x = -\frac{\alpha(t)}{2\eta}. \quad (2)$$

This is a balance of forces. Risk ($\kappa^2 x$) pushes the trader to sell, alpha ($\alpha/2\eta$) pushes the trader to hold, and $\ddot{x}$, the change in trading speed, absorbs the difference. Each discrete condition links only x_{k-1} , x_k and x_{k+1} , so the discrete problem is a tridiagonal linear system.

No Alpha: The Liquidation Curve

With $\alpha = 0$, equation (2) is $\ddot{x} = \kappa^2 x$, solved by $e^{\pm\kappa t}$. The combination with $x(T) = 0$ and $x(0) = X$ is

$$x_{\text{base}}(t) = X \frac{\sinh \kappa(T-t)}{\sinh \kappa T}, \quad (3)$$

the continuous limit of the primer’s schedule, where the discrete decay rate θ tends to κ . $1/\kappa = \sqrt{\eta/\lambda\sigma^2}$ is the time scale of the unwind.

An Earnings Event

Let all the alpha arrive at the print, $\alpha(t) = a\delta(t - t_e)$, so the trader earns $ax(t_e)$ on whatever is held through it. Integrating (2) over a vanishing window around t_e : the integral of $\ddot{x}$ is the change in speed across the window, the integral of $\kappa^2 x$ vanishes, and the delta integrates to a , so

$$\dot{x}(t_e^+) - \dot{x}(t_e^-) = -\frac{a}{2\eta}. \quad (4)$$

The position is continuous, since a jump would need infinite speed and infinite cost. Only the trading rate

jumps: for $a > 0$ the trader sells $a/2\eta$ shares per day faster after the print, so the path has a corner at t_e .

The Green's Function

Let $G(t, s)$ be the response to a unit impulse at time s with both endpoints held at zero: $\partial_t^2 G - \kappa^2 G = -\delta(t - s)$, $G(0, s) = G(T, s) = 0$. Away from s it solves the unforced equation, and gluing the two sides together at s gives

$$G(t, s) = \frac{\sinh(\kappa t_{<}) \sinh \kappa(T - t_{>})}{\kappa \sinh \kappa T}, \quad (5)$$

where $t_{<}$ is the smaller of t and s , and $t_{>}$ the larger. Equation (2) is linear, so solutions add. The base curve (3) carries the boundary conditions and the Green's term carries the event while vanishing at both ends:

$$x(t) = x_{\text{base}}(t) + \frac{a}{2\eta} G(t, t_e). \quad (6)$$

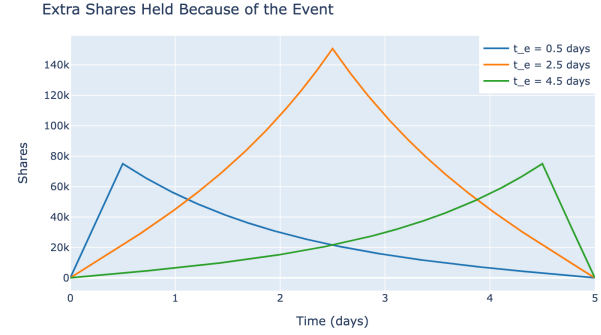
For a general alpha the second term becomes $(2\eta)^{-1} \int_0^T G(t, s) \alpha(s) ds$.

For $a > 0$ the trader holds extra inventory into the print and sells it off faster afterward. For $a < 0$ the trader sells ahead and arrives lighter. With the Almgren-Chriss Table 1 stock ($X = 10^6$ shares, $\sigma = 0.95$, $\eta = 2.5 \times 10^{-6}$, $T = 5$ days), $\lambda = 10^{-6}$ ($\kappa = 0.60$ per day) and a mid-horizon print worth $a = 1$ dollar per share, the trader carries 363,000 shares through it instead of 212,000.



The bump reaches only about $1/\kappa$ either side of the print, so an event far away barely affects trading today. An event near either end of the horizon has little effect,

because G vanishes there and there is no time to build or unwind the extra position.



Event Risk

An earnings print also carries jump variance σ_J^2 . Adding $\lambda \sigma_J^2 x(t_e)^2$ to (1), the jump condition picks up a second term,

$$\dot{x}(t_e^+) - \dot{x}(t_e^-) = \rho x_e - \frac{a}{2\eta}, \quad \rho = \frac{\lambda \sigma_J^2}{\eta}, \quad (7)$$

with $x_e = x(t_e)$. The impulse now depends on the unknown x_e , but superposition still holds:

$$x(t) = x_{\text{base}}(t) + G(t, t_e) \left(\frac{a}{2\eta} - \rho x_e \right).$$

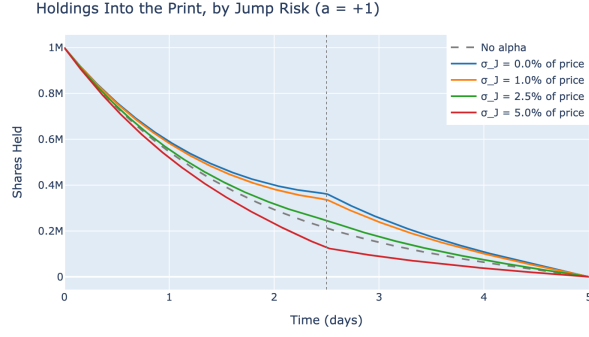
Setting $t = t_e$ and solving for x_e , with $G_{ee} = G(t_e, t_e)$,

$$x_e = \frac{x_{\text{base}}(t_e) + G_{ee} a / 2\eta}{1 + \rho G_{ee}} = w x_{\text{base}}(t_e) + (1 - w) x^*, \quad (8)$$

where

$$w = \frac{1}{1 + \rho G_{ee}}, \quad x^* = \frac{a}{2\lambda \sigma_J^2}.$$

x^* is the position the event alone would call for: minimizing $\lambda \sigma_J^2 x^2 - ax$. The position at the print is a blend of where the liquidation would put the trader and what the event alone wants, and the larger the jump risk, the more the event wins. When $x^* < x_{\text{base}}(t_e)$, the trader arrives lighter than with no view at all, even though the alpha is positive. In the example, a jump of 5% of the price ($\sigma_J = 2.5$ dollars) gives $w = 0.35$ and $x^* = 80,000$, and the position at the print falls to 126,000 shares.



The Book, Translated

Paleologo [2] solves this model for a round trip: start flat, build a position to capture the event alpha, and unwind it ($X = 0$), with no event-variance term. His symbols map into ours as follows.

book	here
$\kappa^2/2$	η
$\rho\sigma^2/2$	$\lambda\sigma^2$
b	κ
α_0, t_0	a, t_e

With these substitutions the book's equation $\kappa^2\ddot{x} = -\alpha + \rho\sigma^2x$ is exactly (2). Its F , G and c solve it by variation of parameters: F and G are the two running integrals of $\alpha e^{\pm b\zeta}$, and c enforces $x(T) = 0$. That is a longer route to the same Green's function. With $X = 0$ the base curve vanishes and only the Green's term of (6) remains.

Long horizon. As $T \rightarrow \infty$, $\sinh \kappa(T - t_>)/\sinh \kappa T \rightarrow e^{-\kappa t_>}$, so

$$x(t) = \frac{a}{2\eta\kappa} \sinh(\kappa t_<) e^{-\kappa t_>},$$

$$x(t_e) = \frac{a}{4\eta\kappa} (1 - e^{-2\kappa t_e}), \quad (9)$$

which are the book's long-horizon results. Before the event the trader buys ever faster, at a rate proportional to $\cosh \kappa t$. After it the position unwinds along $e^{-\kappa t}$, quickly at first and then slowly.

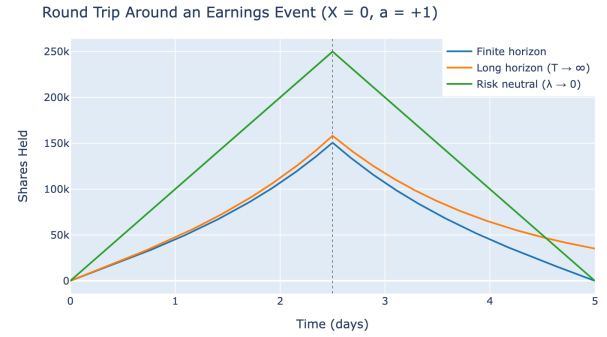
The trading rate at the event. Just before the print the buying rate is $\frac{a}{4\eta}(1 + e^{-2\kappa t_e})$, and just after it the selling rate is $\frac{a}{4\eta}(1 - e^{-2\kappa t_e})$. They differ by exactly the jump $a/2\eta$ of (4). The book says they are equal in size, which holds only when the event is far away ($\kappa t_e \gg 1$).

Risk-aversion convention. The book's problem statement prices risk as $\rho\sigma^2/2$, but its Lagrangian and its rate $b = \rho\sigma/\kappa$ use $\rho^2\sigma^2$. One convention has to be picked. With $\rho\sigma^2$, the rate is $b = \sigma\sqrt{\rho}/\kappa$.

Equation (11.9). The book's risk-neutral solution has two branches that do not meet at t_0 : the left gives $\alpha_0 t_0/\kappa^2$ and the right $\alpha_0 t_0(1 - t_0)/\kappa^2$. It also mixes $T \rightarrow \infty$ with $T = 1$. The limit $\lambda \rightarrow 0$ of (5) is $G \rightarrow t_<(T - t_>)/T$, so the position is a tent, continuous at the print:

$$x(t) = \frac{a}{2\eta} \frac{t_<(T - t_>)}{T}. \quad (10)$$

With $T = 1$ the left branch is $\alpha_0(1 - t_0)t/\kappa^2$, not $\alpha_0 t/\kappa^2$. With $T \rightarrow \infty$ the position after t_0 is flat: a risk-neutral trader never unwinds.



What the book leaves out. The book notes that conviction is strongest on the earnings date, but its risk term uses the same σ everywhere. The position it carries through the print therefore ignores the print's own variance, which is the largest risk in the trade. With $X = 0$, (8) gives

$$x_e = \frac{G_{ee}a/2\eta}{1 + \rho G_{ee}} \rightarrow x^* = \frac{a}{2\lambda\sigma_J^2}$$

as $\sigma_J \rightarrow \infty$: the event variance caps the book's position at the event-only size, $\alpha_0/\rho\sigma_J^2$ in the book's units.

Volume time. The book measures time in volume and keeps σ and κ constant on that clock. That is an assumption about clock time: both are constant in volume time only if variance per unit clock time grows with volume, $\sigma_t^2 \propto v_t$, and impact shrinks with it, $\eta_t \propto 1/v_t$, so that $\sigma_t^2\eta_t$ is constant (appendix).

Appendix

Optimality Equation in Continuous Time

Equation (2)

Suppose $x(t)$ minimizes (1), and perturb it to $x + \varepsilon h$, where $h(0) = h(T) = 0$ so the endpoints stay fixed. At the minimum no perturbation lowers J to first order, so

$$\int_0^T [2\eta \dot{x} \dot{h} + 2\lambda \sigma^2 x h - \alpha h] dt = 0 \quad \text{for every } h.$$

Integrating the first term by parts, and using $h(0) = h(T) = 0$ to drop the boundary term, $\int \dot{x} \dot{h} dt = -\int \ddot{x} h dt$, so

$$\int_0^T [-2\eta \ddot{x} + 2\lambda \sigma^2 x - \alpha] h dt = 0 \quad \text{for every } h.$$

An integral that vanishes against every h forces the bracket to vanish. Dividing by -2η gives equation (2). Integration by parts plays the role of the discrete observation that x_k appears in two neighboring trades.

Jump Conditions

Equations (4) and (7)

Write the event alpha as $a\delta(t - t_e)$ and include the event variance $\lambda\sigma_J^2 x(t_e)^2$. $\dot{x}$ may jump at t_e , so integrate by parts separately on $[0, t_e]$ and $[t_e, T]$:

$$\int_0^T \dot{x} \dot{h} dt = h(t_e)[\dot{x}(t_e^-) - \dot{x}(t_e^+)] - \int_0^T \ddot{x} h dt.$$

The first-order change in J is then

$$\int_0^T [-2\eta \ddot{x} + 2\lambda \sigma^2 x] h dt + h(t_e)[2\eta(\dot{x}(t_e^-) - \dot{x}(t_e^+)) + 2\lambda \sigma_J^2 x_e - a].$$

Perturbations that vanish at t_e give equation (2) away from the event. A perturbation with $h(t_e) \neq 0$ then forces the second bracket to vanish:

$$\dot{x}(t_e^+) - \dot{x}(t_e^-) = \frac{\lambda \sigma_J^2}{\eta} x_e - \frac{a}{2\eta} = \rho x_e - \frac{a}{2\eta},$$

which is (7), and (4) when $\sigma_J = 0$.

Green's Function

Equation (5)

Away from s , G solves $\partial_t^2 G = \kappa^2 G$. On each side, take the solution that already satisfies that side's boundary condition:

$$G = A \sinh \kappa t \quad (t < s), \quad G = B \sinh \kappa(T - t) \quad (t > s).$$

Two conditions at s fix A and B . G is continuous,

$$A \sinh \kappa s = B \sinh \kappa(T - s),$$

and, by (4) with a unit impulse, its slope drops by 1,

$$-B\kappa \cosh \kappa(T - s) - A\kappa \cosh \kappa s = -1.$$

Substituting $B = A \sinh \kappa s / \sinh \kappa(T - s)$ into the second condition,

$$A\kappa(\sinh \kappa s \cosh \kappa(T-s) + \cosh \kappa s \sinh \kappa(T-s)) = \sinh \kappa(T-s).$$

The bracket is $\sinh(\kappa s + \kappa(T-s)) = \sinh \kappa T$, so

$$A = \frac{\sinh \kappa(T-s)}{\kappa \sinh \kappa T}, \quad B = \frac{\sinh \kappa s}{\kappa \sinh \kappa T},$$

and both branches combine into equation (5).

Long-Horizon Limit

Equation (9)

For fixed $t_>$, as $T \rightarrow \infty$,

$$\frac{\sinh \kappa(T-t_>)}{\sinh \kappa T} = \frac{e^{-\kappa t_>} - e^{-\kappa(2T-t_>)}}{1 - e^{-2\kappa T}} \rightarrow e^{-\kappa t_>},$$

which turns (6) with $X = 0$ into the first line of (9). At $t = t_e$, $\sinh(\kappa t_e)e^{-\kappa t_e} = \frac{1}{2}(1 - e^{-2\kappa t_e})$ gives the second. Differentiating, the rate just before the print is

$$\dot{x}(t_e^-) = \frac{a}{2\eta} \cosh(\kappa t_e)e^{-\kappa t_e} = \frac{a}{4\eta}(1 + e^{-2\kappa t_e}),$$

and just after it

$$\dot{x}(t_e^+) = -\frac{a}{2\eta} \sinh(\kappa t_e)e^{-\kappa t_e} = -\frac{a}{4\eta}(1 - e^{-2\kappa t_e}).$$

Their difference is $-a/2\eta$, as (4) requires.

Risk-Neutral Limit

Equation (10)

As $\kappa \rightarrow 0$, $\sinh z \approx z$ in each factor of (5):

$$G(t, s) \approx \frac{\kappa t_< \cdot \kappa(T-t_>)}{\kappa \cdot \kappa T} = \frac{t_<(T-t_>)}{T}.$$

Both branches equal $t_e(T-t_e)/T$ at $t = s = t_e$, so the tent is continuous, and its slopes, $(T-t_e)/T$ before and $-t_e/T$ after, differ by exactly -1 .

Volume Time

Let v_t be the trading volume at clock time t , V its average, and u volume time, $du = (v_t/V) dt$. Over dt the price variance is $\sigma_t^2 dt = \sigma_t^2(V/v_t) du$, so the variance per unit volume time is $\sigma_t^2 V/v_t$. The impact cost is $\eta_t(dx/dt)^2 dt = \eta_t(v_t/V)(dx/du)^2 du$, so the impact per unit volume time is $\eta_t v_t/V$. Both are constant in u only if $\sigma_t^2 \propto v_t$ and $\eta_t \propto 1/v_t$, which makes $\sigma_t^2 \eta_t$ constant in clock time.

Notation

symbol	meaning	units
$x(t)$	shares held at time t	shares
X	starting position ($X = 0$ for a round trip)	shares
T	trading horizon	day
η	temporary impact coefficient	\$ day/share ²
σ	price volatility	\$/ (share day ^{1/2})
λ	risk aversion	1/\$

symbol	meaning	units
κ	liquidation rate, $\sqrt{\lambda\sigma^2/\eta}$	1/day
$\alpha(t)$	expected return from holding	\$/ (share day)
a	alpha paid at the event	\$/share
t_e	event time	day
σ_J	jump volatility at the event	\$/share
ρ	event-risk rate, $\lambda\sigma_J^2/\eta$	1/day
$G(t, s)$	Green's function of (2)	day
x_e	position at the event, $x(t_e)$	shares
x^*	event-only position, $a/2\lambda\sigma_J^2$	shares

References

- [1] R. Almgren and N. Chriss, "Optimal execution of portfolio transactions," *Journal of Risk*, vol. 3, no. 2, pp. 5–39, 2000.
- [2] G. A. Paleologo, *Advanced Portfolio Management: A Quant's Guide for Fundamental Investors*. Wiley, 2021.